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The influence of tangential mean flow on the Rayleigh conductivity of an aperture

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An investigation is made of the influence of grazing mean flow on the Rayleigh conductivity of an aperture in a thin rigid plane. The fluid is assumed to be incompressible and inviscid, but the Kutta condition is applied to permit the generation of vorticity at the edge of the aperture by an applied time-varying pressure field. Numerical results are given for a circular aperture in the two cases of (i) one-sided mean flow, when the aperture is spanned by a plane vortex sheet in the undisturbed state, and (ii) two-sided mean flow, when the mean velocity is the same on both sides of the plane, so that the undisturbed motion is irrotational. In both cases there exist frequency ranges within which perturbation energy is either absorbed or generated by the mean flow. The numerical results are supplemented by an approximate analytical treatment of the same problem for a rectangular aperture of large aspect ratio (with its long edge transverse to the mean flow direction). The aperture flux for one-sided flow is shown to be absolutely unstable, and may in principle be triggered by an arbitrary, small disturbance in the mean stream. For two-sided flow the motion is conditionally unstable, in the sense that perturbations are amplified by the extraction of energy from the mean flow only when the frequency of the applied pressure lies in certain discrete bands.

1. Introduction

Consider the unsteady flow of fluid through an aperture of area A in a rigid plane of negligible thickness produced by a uniform, time-harmonic pressure load $[p] = (p_+ - p_-)e^{-i\omega t}$ ($x > 0$) applied across the plane. The ‘upper’ and ‘lower’ faces of the plane are taken to coincide respectively with surfaces $z = \pm 0$ of the coordinate plane $z = 0$ of the rectangular system (x, y, z) , and p_+ , p_- are the respective amplitudes of the applied pressures above and below the plane. The induced motion through the aperture is assumed to be *incompressible* and to be governed by the linearized equations of motion. Let Q denote volume flux through the plane (in the positive z -direction), where here and henceforth the exponential time factor $e^{-i\omega t}$ is suppressed. Then the Rayleigh conductivity K_R of the aperture is defined by (Rayleigh 1870, 1945)

$$K_R = i\omega\rho_0 Q / (p_+ - p_-), \quad (1.1)$$

where ρ_0 denotes the density of the fluid.

The pressure varies continuously through the aperture, and $p_{\pm}$ are its limiting values at large distances (greater than $\sqrt{A}$) on either side. When there is no *mean*

flow, and the motion at large distances can be regarded as irrotational, the definition (1.1) is equivalent to

$$K_R = Q/(\varphi_+ - \varphi_-), \quad (1.2)$$

where $\varphi_{\pm}$ are the *uniform* velocity potentials above and below the plane, determined by $p_{\pm} = -\rho_0 \partial \varphi_{\pm} / \partial t$. This form of the definition is analogous to Ohm's law in electricity. Conductivity has the dimensions of length; for example, it is equal to $2R$ for a circular aperture of radius R in an ideal, stationary fluid.

For an ideal, inviscid fluid, K_R is real and the motion through the aperture is entirely irrotational and conservative. In practice, however, viscous action causes dissipation, and the dissipated power Π , say, is given in terms of $[p]$ and the normal component v_n of fluid velocity at $z = 0$ by

$$\Pi = - \int_{-\infty}^{\infty} \langle [p] v_n \rangle \, dx \, dy,$$

where the angle brackets denote a time average, and the real parts of $[p]$, v_n are to be taken after restoration of the time factor $e^{-i\omega t}$. It follows that

$$\Pi = -\frac{1}{2} \operatorname{Im}\{K_R\} |[p]|^2 / \rho_0 \omega, \quad (1.3)$$

and therefore that energy is dissipated provided the imaginary part of K_R is negative.

The dissipation is frequently enhanced for certain frequencies by the presence of a high Reynolds number mean tangential flow over the aperture or a 'bias' flow through the plane (Bechert 1979; Crighton 1981), and perforated screens have been deployed in mean flow environments to eliminate harmful acoustic resonances (for example, in heat exchanger cavities (Vèr 1990)). For a bias flow screen with circular apertures, the dissipation can be estimated from (1.3) by use of an approximate formula for K_R given by Howe (1979), which has been validated experimentally by Hughes & Dowling (1990). The mean flow is maintained by a steady pressure differential across the screen. If U is the average velocity within an aperture, and $U\sqrt{A}/\nu \gg 1$, the motion will be uninfluenced by viscosity ν except in the immediate vicinity of the aperture edge, where the flow separates to form a jet. The undisturbed motion is no longer irrotational, and a superimposed, fluctuating pressure $[p]$ causes additional vorticity to be generated, which is then swept away in the jet. The energy required to produce this vorticity is supplied by the applied pressure $[p]$, and dissipation occurs by its transfer to the kinetic energy of the jet.

In this paper we consider the determination of K_R for an aperture in the presence of a tangential mean flow. When the mean velocities are at speeds U_+ , U_- above and below the plane (see figure 1) and the Reynolds number is large, the undisturbed mean shear layer in the aperture can be approximated by a vortex sheet. The sheet is deformed by a small amplitude applied pressure $[p]$, and additional vorticity is shed from the upstream edge of the aperture. However, because the vortex sheet is unstable (Lamb 1932), perturbation energy may be expected to be either created or absorbed by the interaction, depending on the frequency ω of the applied pressure. This problem was examined by Howe (1980), who obtained high and low frequency approximations for K_R for a circular aperture when $U_+ = U_-$, so that the undisturbed motion may be regarded as irrotational. The applied pressure was assumed to induce separation from the upstream semi-circular edge of the aperture in the form of a thin vortex sheet, the strength of which was set by the Kutta condition (Crighton 1985), and it was concluded that $\operatorname{Im}\{K_R\} < 0$ for all ω . We shall argue

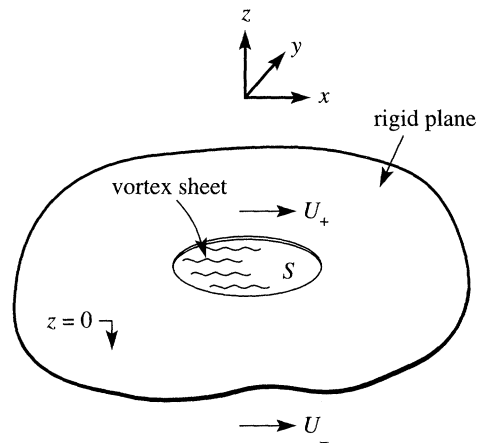


Figure 1. Tangential flow over the upper and lower surfaces of a thin rigid plane with a circular aperture.

below in §2 that Howe's (1980) solution is erroneous, because proper account was not taken of conditions at the downstream edge of the aperture.

The tangential flow problem is formulated in §2, and solved numerically for a circular aperture for the two cases of (i) one-sided mean flow, when the fluid is at rest on one side of the plane in the undisturbed state, and (ii) two-sided flow at the same velocity on both sides. The problem is amenable to approximate analytical treatment for a *rectangular* aperture of large aspect ratio, orientated with its long edges perpendicular to the mean flow direction. This is discussed in §3, and comparison made with the numerical results. In all cases there exist frequency ranges where $\text{Im}\{K_R\} > 0$, within which perturbation energy is extracted from the mean flow. For both the circular and rectangular apertures we show in case (ii), for which the undisturbed motion is irrotational, that the motion is *conditionally* unstable (§4), i.e. that energy is extracted from the mean flow only when $[p] \neq 0$; in case (i), on the other hand, the motion is *absolutely* unstable, and an arbitrarily small disturbance can trigger the release of mean flow kinetic energy.

A brief discussion is given in the Appendix of the *generalized* Rayleigh conductivity $K_R(\mathbf{k}, \omega)$, which is defined as in (1.1), but for an applied pressure $[p] = p_0 e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$, where p_0 is constant and $\mathbf{k}$ is an arbitrary 'trace-wavenumber' parallel to plane. Rayleigh's (1870) definition corresponds to $K_R(\mathbf{0}, \omega)$. The reverse flow reciprocal theorem (Howe 1975; Möhring 1979), extended to problems involving vortex sheets, can be invoked to express $K_R(\mathbf{k}, \omega)$ in terms of the vortex sheet displacement in the reverse flow problem for $\mathbf{k} = \mathbf{0}$.

2. Circular aperture with a grazing mean flow

(a) The vortex sheet model

Consider incompressible, uniform mean flows over the upper and lower surfaces of the rigid plane $z = 0$ which is pierced by a circular aperture of radius R , the centre of which is at the coordinate origin. Let the mean velocities be at speeds $U_{\pm}$ in the x -direction respectively in $z \lesseqgtr 0$, as in figure 1. The Reynolds number is assumed to be sufficiently large that viscosity can be neglected except for its role in generating

vorticity at sharp edges. In the steady state, when $U_+ \neq U_-$, the aperture is spanned by a vortex sheet.

When the plane is subject to a small amplitude, time harmonic pressure load $[p] = p_+ - p_-$, the velocity potentials $\varphi_{\pm}(\mathbf{x})$ of the unsteady motion above and below the plane may be expressed in the form (Lamb 1932)

$$\varphi_{\pm}(\mathbf{x}) = \frac{\mp 1}{2\pi} \int_{-\infty}^{\infty} \frac{v_{\pm}(x', y')}{|\mathbf{x} - \mathbf{x}'|} dx' dy', \quad \mathbf{x}' = (x', y', 0), \quad (2.1)$$

where $v_{\pm}$ is the z -component of velocity as $z \rightarrow \pm 0$, which vanishes on the plane except at the aperture. In the linearized approximation, the z -component of displacement $\zeta(x, y)$ of fluid particles initially lying just above or below the plane ($z = \pm 0$) vanishes outside the aperture, and equals the displacement of the vortex sheet within the aperture. Thus

$$v_{\pm}(x, y) = (\partial/\partial t + U_{\pm}\partial/\partial x)\zeta \equiv -i(\omega + iU_{\pm}\partial/\partial x)\zeta, \quad (2.2)$$

and integration by parts permits (2.1) to be cast in the form

$$\varphi_{\pm}(\mathbf{x}) = \frac{\pm i}{2\pi} \left(\omega + iU_{\pm} \frac{\partial}{\partial x} \right) \int_S \frac{\zeta(x', y')}{|\mathbf{x} - \mathbf{x}'|} dx' dy', \quad (2.3)$$

where the final integration may be restricted to the area S occupied by aperture provided $\zeta \sim O(1/|X|^{\alpha})$, $\alpha < 1$, near the aperture edge, where X denotes distance from the edge.

The volume flux Q through the aperture is needed to calculate the conductivity K_R from the definition (1.1). This is given by

$$Q = \int_{-\infty}^{\infty} -i(\omega + iU_{\pm}\partial/\partial x)\zeta dx dy \equiv -i\omega \int_S \zeta(x, y) dx dy. \quad (2.4)$$

The equation determining $\zeta(x, y)$ in terms of $[p]$ is obtained from the condition that the pressures just above and below the vortex sheet must be equal. The linearized Bernoulli equation supplies the perturbation pressure in the form

$$p_{\pm} + i\rho_0(\omega + iU_{\pm}\partial/\partial x)\varphi_{\pm} \quad (z \lesssim 0).$$

Continuity of pressure across the vortex sheet therefore requires that

$$\begin{aligned} \left[\left(\omega + iU_+ \frac{\partial}{\partial x} \right)^2 + \left(\omega + iU_- \frac{\partial}{\partial x} \right)^2 \right] \int_S \frac{\zeta(x', y')}{2\pi|\mathbf{x} - \mathbf{x}'|} dx' dy' \\ = (p_+ - p_-)/\rho_0, \quad \text{for } z = 0, \quad (x^2 + y^2)^{1/2} < R. \end{aligned} \quad (2.5)$$

Integrating with respect to the second order differential operator on the left-hand side, we find (for constant $(p_+ - p_-)/\rho_0$)

$$\int_S \frac{Z(\xi', \eta') d\xi' d\eta'}{\sqrt{(\xi - \xi')^2 + (\eta - \eta')^2}} = 1 + \alpha_1(\eta)e^{i\sigma_1\eta} + \alpha_2(\eta)e^{i\sigma_2\eta}, \quad (\eta^2 + \xi^2)^{1/2} < 1, \quad (2.6)$$

where $Z = \{\rho_0\omega^2 R/\pi(p_+ - p_-)\}\zeta$ is the displacement in dimensionless form, $\xi = x/R$, $\eta = y/R$, etc. (so that the integration is over the unit disk $(\xi'^2 + \eta'^2)^{1/2} < 1$), and σ_1 and σ_2 are the dimensionless Kelvin–Helmholtz wavenumbers of instability waves on the vortex sheet (Lamb 1932)

$$\sigma_1 = \frac{\omega R(1 + i)}{U_+ + iU_-}, \quad \sigma_2 = \frac{\omega R(1 - i)}{U_+ - iU_-}. \quad (2.7)$$

The amplitudes $\alpha_1(\eta)$, $\alpha_2(\eta)$ of these waves are constant along any line drawn on the sheet in the streamwise direction, but vary with transverse location η . They must be chosen (in accordance with the Kutta condition) to ensure that the vortex sheet leaves the upstream semi-circular arc of the aperture edge tangentially, i.e. such that $Z(\xi, \eta) = \partial Z(\xi, \eta)/\partial \xi = 0$ for $(\xi^2 + \eta^2)^{1/2} = 1$, $\xi < 0$. Evidently, $\alpha_1(\eta)$, $\alpha_2(\eta)$ are even functions of η . Potential theory (Kellogg 1954; Sneddon 1966) implies that ζ will generally exhibit a mild singularity of order $1/\sqrt{R-r}$ at the downstream edge $r \equiv (x^2 + y^2)^{1/2} = R$, $x > 0$, of the aperture. The singularity is integrable, however (yielding a finite value for the volume flux Q (2.4)), and is a physically acceptable, linear theory representation of the large amplitude and violent motions produced by vortex impingement on an edge (Rockwell 1983). A correction that incorporates the influence of displacement thickness waves on the downstream surfaces of the plane (modelling vorticity convected in the boundary layers after ejection from the aperture) can be introduced to eliminate the singularity, but this cannot be done in a unique manner and is not expected to modify significantly the conclusions of the present simpler approach (Howe 1981*b*).

The solution of the integral equation (2.6) determines the volume flux Q via (2.4). The definition (1.1) then yields

$$K_R = \pi R \int_S Z(\xi, \eta) d\xi d\eta. \quad (2.8)$$

In the absence of flow the conductivity is real and equal to $2R$. In the more general case it is convenient to set

$$K_R = 2R(\Gamma_R - i\Delta_R), \quad (2.9)$$

where Γ_R and Δ_R are real valued and depend on the frequency ω .

(*b*) Numerical results

Numerical solutions of equation (2.6) have been obtained by discretizing the integral. The aperture was overlaid by a mesh of square integration elements within each of which Z , α_1 and α_2 were assumed to be constant, and the Kutta condition imposed by requiring Z to vanish in those elements overlapping the upstream edge (where the flow separates) *and* in each of the contiguous elements just downstream. The application of (2.6) in each element of the mesh supplies a sufficient number of equations to determine the discretized values of Z , α_1 and α_2 . Further details of the procedure are given by Scott (1995). In the degenerate case in which the mean velocities on each side of the plane are the same and equal to U (so that $\sigma_1 = \sigma_2 = \omega R/U$), the right-hand side of equation (2.6) was taken in the modified form

$$1 + (\alpha_1(\eta) + \alpha_2(\eta)\xi) e^{i\sigma\xi}, \quad \text{where } \sigma = \omega R/U.$$

The real and imaginary parts of the conductivity (2.9) are functions of the Strouhal number $\omega R/U_+$ and the velocity ratio U_-/U_+ . Their calculated values are given in tables 1 and 2 for the two cases of: (i) one-sided flow, $U_+ = U$, $U_- \equiv 0$; and (ii) two-sided flow, $U_+ = U_- = U$. In applications these tabulated results can be approximated by cubic splines, and this has been done in plotting Γ_R and Δ_R for these two cases in figure 2.

For one-sided flow (case (i)) $\Gamma_R \approx 4$ when $\omega R/U$ becomes large and the imaginary part $\Delta_R \rightarrow 0$. Δ_R is negative for $1.9 < \omega R/U < 3.9$. According to equation (1.3), a negative amount of work is then done by the applied pressure $[p] = p_+ - p_-$,

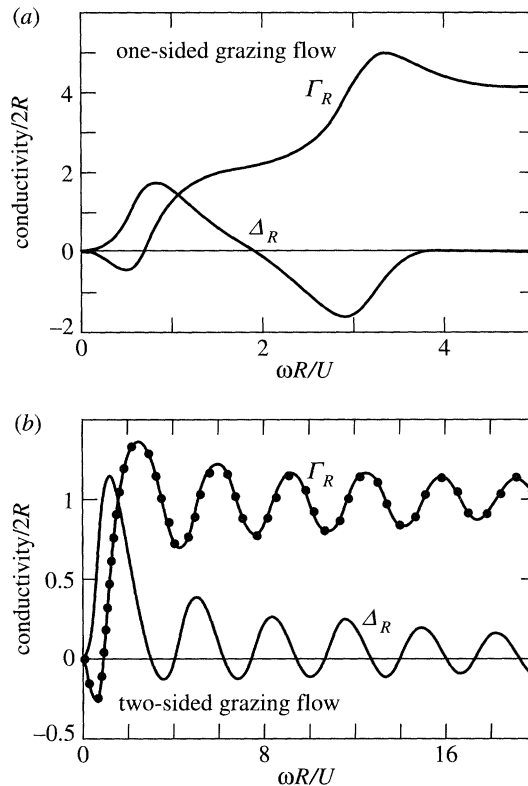


Figure 2. Conductivity of a circular aperture in grazing flow: (a) $U_+ = U$, $U_- = 0$; (b) $U_+ = U_- = U$. The dotted curve in (b) is discussed in § 4.

i.e. energy is extracted from the mean flow. In case (ii) ($U_+ = U_-$) the motion is irrotational when $[p] = 0$. However, there still exist periodically spaced frequency bands (of period π when $\omega R/U > 3$) within which $\Delta_R < 0$, and where perturbations grow at the expense of the mean flow. This extraction of mean flow energy is not associated with vortex sheet instability, but should in this case be regarded as a consequence of the interaction of the vorticity with the semi-circular ‘leading edge’ on which it impinges after convecting across the aperture. We shall show in § 4 that cases (i) and (ii) are fundamentally different, inasmuch as the aperture motions are respectively absolutely and conditionally unstable.

(c) Comparison with previous work

The two-sided flow problem where $U_+ = U_- = U$ has been investigated by Howe (1980), who derived the following integral equation for the dimensionless normal velocity $v' = -i(\sigma + i\partial/\partial\xi)Z$ of the vortex sheet

$$\int_S \frac{v'(\xi', \eta') d\xi' d\eta'}{\sqrt{(\xi - \xi')^2 + (\eta - \eta')^2}} = C + \alpha(\eta)e^{i\sigma\xi}, \quad (\xi^2 + \eta^2)^{1/2} < 1, \quad (2.10)$$

where C is a known constant. The Kutta condition was imposed by choosing $\alpha(\eta)$ to ensure that v' is finite over the upstream edge $(\xi^2 + \eta^2)^{1/2} = 1$, $\xi < 0$ of the aperture. Consequently, $v \sim O(1/\sqrt{R-r})$ at the downstream edge, on which the shed vorticity impinges after convecting across the aperture. In fact, equation (2.10)

Table 1. Conductivity of a circular aperture for one-sided grazing flow $U_+ = U$, $U_- = 0$

$\omega R/U$	Γ_R	Δ_R	$\omega R/U$	Γ_R	Δ_R
0.00	0.000	0.000	1.50	1.949	0.623
0.10	-0.030	0.005	1.75	2.097	0.244
0.20	-0.119	0.041	2.00	2.233	-0.131
0.30	-0.253	0.145	2.25	2.409	-0.540
0.40	-0.395	0.360	2.50	2.708	-1.006
0.50	-0.464	0.710	2.75	3.277	-1.459
0.60	-0.350	1.140	3.00	4.184	-1.555
0.70	-0.016	1.503	3.25	4.887	-0.969
0.80	0.439	1.673	3.50	4.897	-0.300
0.90	0.872	1.652	3.75	4.628	-0.000
1.00	1.216	1.515	4.00	4.398	0.059
1.10	1.466	1.333	4.25	4.258	0.033
1.20	1.644	1.142	4.50	4.189	0.003
1.30	1.774	0.959	4.75	4.175	0.011
1.40	1.872	0.786	5.00	4.182	0.115

is invalid because the range of integration in (2.10) cannot be restricted to the region S of the aperture unless account is taken of a *singular* contribution to v' from the downstream edge. When $v \sim O(1/\sqrt{R-r})$ at this edge, the displacement ζ will have a simple discontinuity there, with limiting value ζ_D , say, as $r \rightarrow R-0$ ($x > 0$), and the singular velocity at the edge would equal $-U\zeta_D\delta(r-R)$. This should appear as a separate contribution in (2.10) and must be included in the evaluation of the volume flux Q . A finite limiting behaviour of ζ at the downstream edge is possible only when $U_+ = U_-$, since in general the edge singularity is much stronger than a simple discontinuity, being proportional to $1/\sqrt{R-r}$, although the aperture flux remains finite.

3. The rectangular aperture

It is of interest to compare the numerical predictions for a circular aperture with corresponding results that can be derived in analytical form for a rectangular aperture of large aspect ratio.

Take the mean flow in the x -direction with velocities $U_{\pm}$ above and below the plate, as before, and let the rectangular aperture occupy the region $|x| < s$, $|y| < \frac{1}{2}b$, where the *aspect ratio* $b/2s \gg 1$ (figure 3). In the notation of §2, the vortex sheet displacement ζ satisfies the integro-differential equation (2.5), in which S now denotes the rectangular area. An approximate solution of this equation can be obtained when $b/2s$ is large by assuming the motion of the vortex sheet to be two-dimensional, i.e. by neglecting the dependence of $\zeta(x', y')$ on the spanwise coordinate y' . The integration with respect to y' over $(-\frac{1}{2}b, \frac{1}{2}b)$ in (2.5) may then be performed explicitly. If we also integrate over $-\frac{1}{2}b < y < \frac{1}{2}b$, the resulting equation may be regarded as satisfying the condition of pressure continuity across the vortex sheet in a spanwise averaged sense.

Table 2. *Conductivity of a circular aperture for two-sided grazing flow $U_+ = U_- = U$*

$\omega R/U$	Γ_R	Δ_R	$\omega R/U$	Γ_R	Δ_R
0.00	0.000	0.000	4.00	0.749	-0.031
0.10	-0.015	0.002	4.25	0.691	0.102
0.20	-0.057	0.019	4.50	0.714	0.242
0.30	-0.119	0.063	4.75	0.805	0.345
0.40	-0.187	0.143	5.00	0.928	0.385
0.50	-0.243	0.264	5.25	1.048	0.363
0.60	-0.268	0.420	5.50	1.141	0.297
0.70	-0.247	0.594	5.75	1.195	0.203
0.80	-0.174	0.767	6.00	1.208	0.099
0.90	-0.055	0.916	6.25	1.180	0.002
1.00	0.096	1.030	6.50	1.115	-0.076
1.10	0.263	1.101	6.75	1.023	-0.120
1.20	0.431	1.133	7.00	0.920	-0.117
1.30	0.590	1.131	7.25	0.830	-0.065
1.40	0.734	1.103	7.50	0.778	0.026
1.50	0.860	1.057	7.75	0.781	0.130
1.75	1.102	0.894	8.00	0.836	0.215
2.00	1.252	0.705	8.25	0.922	0.259
2.25	1.331	0.514	8.50	1.014	0.256
2.50	1.354	0.333	8.75	1.091	0.213
2.75	1.329	0.169	9.00	1.139	0.143
3.00	1.260	0.029	9.25	1.152	0.062
3.25	1.152	-0.074	9.50	1.130	-0.017
3.50	1.015	-0.127	9.75	1.075	-0.079
3.75	0.870	-0.114	10.00	0.998	-0.112

To do this observe that, when x and x' both lie within the aperture, so that $|x - x'| < 2s$,

$$\begin{aligned}
 & \int_{-b/2}^{b/2} \int_{-b/2}^{b/2} \frac{dy' dy}{\sqrt{(x - x')^2 + (y - y')^2}} \\
 &= 2b \sinh^{-1}(b/|x - x'|) - 2\{\sqrt{b^2 + (x - x')^2} - |x - x'|\} \\
 &= 2b \left(\ln\{2b/e|x - x'|\} + \frac{|x - x'|}{b} - \frac{|x - x'|^2}{4b^2} + \dots \right), \quad (3.1)
 \end{aligned}$$

where e is the exponential constant. When $b/2s \gg 1$ only the first term in the brace brackets of this expansion need be retained. Making this approximation in (2.5), and integrating with respect to the differential operator, we obtain

$$\int_{-1}^1 Z(\xi') \ln \left(\frac{2b}{es|\xi - \xi'|} \right) d\xi' = 1 + \alpha_1 e^{i\sigma_1 \xi} + \alpha_2 e^{i\sigma_2 \xi}, \quad |\xi| < 1, \quad (3.2)$$

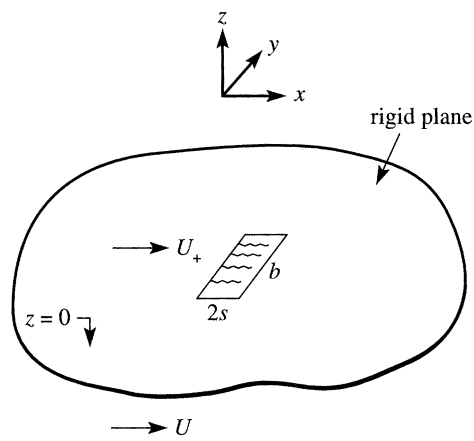


Figure 3. Rectangular aperture of large aspect ratio. The long edges of the aperture are transverse to the mean flow direction.

where

$$Z = \{2\rho_0\omega^2 s/\pi(p_+ - p_-)\}\zeta, \quad \xi = x/s, \quad \xi' = x'/s$$

and σ_1 and σ_2 are nondimensional Kelvin–Helmholtz wavenumbers defined as in (2.7) with R replaced by s . The coefficients α_1 and α_2 are *constants*, the values of which are fixed by the Kutta condition $Z = \partial Z/\partial \xi = 0$ at $\xi = -1$.

The solution of (3.2) satisfying these conditions has been given by Howe (1981*b*). Z varies like $(\xi + 1)^{3/2}$ near the upstream edge $\xi \rightarrow -1 + 0$ of the aperture, and is singular like $1/(1 - \xi)^{1/2}$ at the downstream edge ($\xi \rightarrow 1 - 0$) where the vortex sheet impinges on the solid surface. The volume flux

$$Q = \{\pi b(p_+ - p_-)/2i\rho_0\omega\} \int_{-1}^1 Z(\xi) d\xi,$$

and the Rayleigh conductivity is therefore equal to $\frac{1}{2}\pi \int_{-1}^1 Z(\xi) d\xi$. Using the solution given by Howe (1981*b*) we find

$$K_R = \frac{\pi b}{2\{F(\sigma_1, \sigma_2) + \Psi\}}, \quad (3.3)$$

where

$$F = \frac{-\sigma_1 J_0(\sigma_2)[J_0(\sigma_1) - 2W(\sigma_1)] + \sigma_2 J_0(\sigma_1)[J_0(\sigma_2) - 2W(\sigma_2)]}{\sigma_1 W(\sigma_2)[J_0(\sigma_1) - 2W(\sigma_1)] - \sigma_2 W(\sigma_1)[J_0(\sigma_2) - 2W(\sigma_2)]}, \quad \Psi = \ln(4b/es), \quad (3.4)$$

$W(x) = ix[J_0(x) - iJ_1(x)]$, and J_0 and J_1 are Bessel functions.

The conductivity in the absence of flow is also given by (3.3) by formally setting $F = 0$. For that case we can write

$$K_R/2R_e = (\pi^3 b/32s)^{1/2}/\ln\{4b/se\}, \quad (3.5)$$

where $R_e = (2sb/\pi)^{1/2}$ is the radius of the circle of area $2sb$ (that of the rectangle). According to Rayleigh (1945) the circular aperture has the minimum conductivity of all apertures of the same area, and indeed the right-hand side of (3.5) has a minimum value of about 1.1, and does not exceed 1.2 when $2.5 < b/s < 12.8$.

Figure 4*a* illustrates the dependence of the real and imaginary parts of

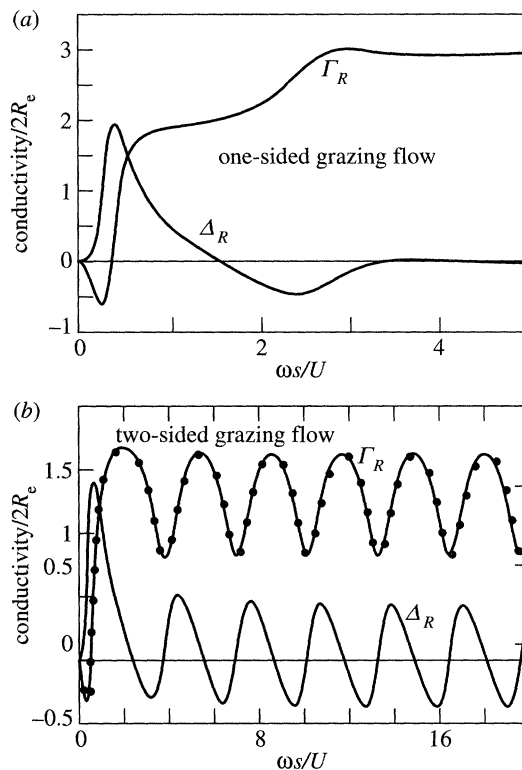


Figure 4. Conductivity $K_R(\omega)/2R_e = \Gamma_R - i\Delta_R$ of a rectangular aperture of aspect ratio $b/2s = 10$ in grazing flow: (a) $U_+ = U$, $U_- = 0$; (b) $U_+ = U_- = U$. The dotted curve in (b) is discussed in § 4.

$K_R(\omega)/2R_e = \Gamma_R - i\Delta_R$ on $\sigma = \omega s/U$ for one sided flow, $U_+ = U$, $U_- = 0$ when the aspect ratio $b/2s = 10$. These curves are qualitatively the same as the corresponding plots in figure 2a for the circular aperture. It follows from equations (3.3), (3.4) for the rectangular aperture that

$$\left. \begin{aligned} \Gamma_R &\approx \frac{(\pi^3 b/32s)^{1/2}}{\ln(4b/se^3)}, \quad \Delta_R \approx 0, \quad \sigma \rightarrow \infty, \\ \Gamma_R &\approx -3\sigma^2(\pi^3 b/32s)^{1/2}, \quad \Delta_R \approx 8\sigma^3(\pi^3 b/32s)^{1/2}, \quad \sigma \rightarrow 0. \end{aligned} \right\} \quad (3.6)$$

Thus, at high frequencies the conductivity is real and exceeds that in the absence of the vortex sheet (provided $4b/se^3 > 1$).

For two-sided flow with $U_+ = U_- = U$, there is no vortex sheet in the steady state and $\sigma_1 \rightarrow \sigma_2 \rightarrow \sigma = \omega s/U$. The corresponding limiting form of function F of (3.4) is

$$F(\sigma, \sigma) = \frac{i\sigma(J_0 - iJ_1) - [J_0 - 2i\sigma(J_0 - iJ_1)][J_0 - i\sigma(J_0 + iJ_1)]}{\sigma[J_0J_1 + \sigma\{J_1^2 + (J_0 - 2iJ_1)^2\}]}, \quad (3.7)$$

where the Bessel functions are evaluated at $\sigma = \omega s/U$. The variations of Γ_R and Δ_R with σ are shown in figure 4b for $b/2s = 10$. Once again, the results are qualitatively similar to those in figure 2b for the circular aperture. However, whereas figure 2b indicates that the oscillatory behaviour of K_R ultimately dies out, and that $\Gamma_R \rightarrow 1$, $\Delta_R \rightarrow 0$ as $\sigma \rightarrow \infty$, the limiting behaviours for the rectangular aperture remain

oscillatory, with period π , since $F \approx 4/\{5 \sin(2\sigma) + 4i \cos(2\sigma) - 3\}$ when $\sigma > 2$. As $\sigma \rightarrow 0$ we find

$$\Gamma_R \approx -\frac{3}{2}\sigma^2(\pi^3 b/32s)^{1/2}, \quad \Delta_R \approx 6\sigma^3(\pi^3 b/32s)^{1/2}. \quad (3.8)$$

4. Absolute and conditional instability

The volume flux Q is determined by the applied pressure $[p] \equiv p_+ - p_-$ and the conductivity K_R . All of these quantities are generally functions of the frequency ω , and the real form of (1.1) is

$$\rho_0 \partial Q(t)/\partial t = - \int_{-\infty}^{\infty} K_R(\omega) [p(\omega)] e^{-i\omega t} d\omega. \quad (4.1)$$

The integration is nominally taken over all real values of ω . However, if the integrand has singularities in the upper half of the complex ω -plane, the integration contour must pass above them to ensure that causality is satisfied. Since $[p(t)]$ is arbitrary, and may be assumed to vanish prior to some initial instant, any singularities in $\text{Im}(\omega) > 0$ must be associated with $K_R(\omega)$. When they are present the system is unstable, and an unsteady flux through the aperture can develop spontaneously. Also, because $Q(t)$ and $[p(t)]$ are real valued quantities, it follows that $K_R(-\omega^*) = K_R^*(\omega)$, where the asterisk denotes complex conjugate.

Consider the stable case in which $K_R(\omega) \equiv 2R\{\Gamma_R(\omega) - i\Delta_R(\omega)\}$, is regular in $\text{Im}(\omega) > 0$. For *real* ω we have

$$\Gamma_R(-\omega) = \Gamma_R(\omega), \quad \Delta_R(-\omega) = -\Delta_R(\omega). \quad (4.2)$$

In general $\Gamma_R(\omega) \rightarrow \Gamma_\infty = \text{constant}$ as $|\omega| \rightarrow \infty$ and $\Delta_R(\omega) \rightarrow 0$. The function $f(\omega) = \Gamma_R(\omega) - \Gamma_\infty - i\Delta_R(\omega)$ is regular in $\text{Im}(\omega) > 0$ and vanishes as $|\omega| \rightarrow \infty$. Cauchy's theorem (Carrier *et al.* 1966) applied in the usual way to a closed contour consisting of the real axis and a semi-circle in the upper half-plane, the radius of which grows without limit, then implies that

$$f(\omega) = (1/2\pi i) \int_{-\infty}^{\infty} f(\xi) d\xi / (\xi - \omega) \quad \text{for } \text{Im}(\omega) > 0,$$

where the integration is along the real axis. The real and imaginary parts of this expression as $\text{Im}(\omega) \rightarrow +0$ yield the Kramers–Kronig formulae

$$\Gamma_R(\omega) - \Gamma_\infty = \frac{2}{\pi} \oint_0^\infty \frac{\lambda \Delta_R(\lambda) d\lambda}{\lambda^2 - \omega^2}, \quad \Delta_R(\omega) = \frac{2\omega}{\pi} \oint_0^\infty \frac{[\Gamma_R(\lambda) - \Gamma_\infty] d\lambda}{\lambda^2 - \omega^2}, \quad (\omega \text{ real}), \quad (4.3)$$

where the integrals are principal values.

When the motion is stable these equations can be used to determine Γ_R , say, when $\Delta_R(\omega)$ is known. If $K_R(\omega)$ vanishes at zero frequency (as it would when vortex shedding completely blocks the aperture flux), the constant Γ_∞ is found by setting $\omega = 0$ in the first of (4.3).

Alternatively, when $\Gamma_R(\omega)$ and $\Delta_R(\omega)$ are known for real ω , equations (4.3) can be used to investigate the stability of the flow. If the equations are not satisfied $K_R(\omega)$ must be singular in $\text{Im}(\omega) > 0$, and the motion is *absolutely unstable*. This is the case for one-sided grazing flow over the circular and rectangular apertures of §§3 and 4, and is presumably a consequence of vortex sheet instability. According

to figures 2*b* and 4*b*, $\Delta_R < 0$ for discrete ranges of real frequencies for two-sided flow when $U_+ = U_- = U$. Equation (1.3) indicates that energy is extracted from the mean flow at these frequencies. However, the Kramers–Kronig formulae are satisfied by the numerical data for two-sided flow in figures 2*b* and 4*b*. The dots in these figures represent values of Γ_R computed from the first of the Kramers–Kronig equations (4.3) using the respective values of Δ_R shown in the figures (taking $\Delta_R(\lambda) = 0$ for $\lambda > 20$). The motion is therefore *conditionally* unstable: energy is extracted from the mean flow only when the applied pressure $[p]$ contains a component with frequency within an interval where $\Delta_R < 0$.

When the aperture is backed by a cavity (in which $U_- \equiv 0$), the dimensions of which are large compared to the aperture diameter, the instability of the shear layer can excite and sustain cavity acoustic resonances. This can occur only when the frequency lies within the range where $\Delta_R(\omega) < 0$. For the circular aperture it is usual to express the frequency at which a resonance is excited in terms of the Strouhal number fD/U , where $f = \omega/2\pi$ and D is the diameter. According to figure 2*a* flow excited cavity resonances must occur for $0.6 < fD/U < 1.15$. This range is consistent with experimentally observed Strouhal numbers for laminar, high Reynolds number flow over cavity openings in cases where the shear layer is well approximated by a vortex sheet, i.e. when the shear layer thickness is very much smaller than D (see, for example, DeMetz & Farabee 1977).

5. Conclusion

The Rayleigh conductivity K_R of an aperture in a thin plane determines the volume flux through the aperture generated by a small amplitude, time harmonic pressure load. When the motion is incompressible and irrotational K_R is real and constant, and depends only on the shape of the aperture. The conductivity is complex for a real fluid. If there is a tangential flow over the plane, and the Reynolds number based on aperture diameter is large, the pressure load can be assumed to generate vorticity at the aperture edge, which causes perturbation energy to be absorbed or generated by the mean flow, according as $\text{Im}\{K_R\} \lesseqgtr 0$.

In this paper K_R has been calculated numerically for tangential flow over a circular aperture of radius R by modelling the shear layer as a vortex sheet. For one-sided flow at speed U , energy is extracted from the mean flow when the Strouhal number $\omega R/U$ is between about 1.9 and 3.9. The aperture motion is absolutely unstable, in the sense that a small perturbation in the mean stream can trigger motion of the vortex sheet that becomes unbounded on the basis of linear theory. When the mean velocity is the same on both sides of the plane the flow is irrotational in the steady state. Vorticity is generated by an applied pressure (in accordance with the Kutta condition) and there are discrete, quasi-periodic frequency intervals within which perturbation energy is alternately absorbed and generated by the mean flow. However, this motion is conditionally unstable, in that perturbation energy grows at the expense of the mean flow only when the applied pressure is non-zero and of the appropriate frequency.

The numerical results for a circular aperture are in qualitative agreement with analytical predictions for a rectangular aperture of large aspect ratio, the long edge of which is transverse to the mean flow direction.

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Appendix A. Generalized Rayleigh conductivity

It is often necessary to consider the interaction of a time harmonic load $[p] = (p_+ - p_-)e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$ with an aperture, where $\mathbf{k} = (k_x, k_y)$ is the ‘trace wavenumber’ of the pressure field over the plane. This occurs in problems concerned with the diffraction of sound by apertures, and also in modelling the interaction of ‘hydrodynamic’ disturbances with an aperture. It is then useful to extend the definition (1.1) to include the *generalized* Rayleigh conductivity $K_R(\mathbf{k}, x)$:

$$K_R(\mathbf{k}, x) = i\omega\rho_0 Q / (p_+ - p_-), \quad [p] = (p_+ - p_-)e^{i\mathbf{k} \cdot \mathbf{x}}. \quad (\text{A } 1)$$

For an ideal incompressible fluid in the absence of mean flow

$$K_R(\mathbf{k}, \omega) = (2/k) \sin(kR)$$

for a circular aperture of radius R (Sneddon 1966). Similarly, for the large aspect ratio rectangular aperture of §3 it can be shown that,

$$K_R(k_x, 0, \omega) = \pi b J_0(k_x s) / 2 \Psi,$$

where Ψ is defined as in (3.4).

In the presence of a tangential flow the generalized conductivity for the large aspect ratio rectangular aperture can be determined in analytic closed form when $k_y = 0$. The non-dimensional displacement Z of the vortex sheet satisfies equation (3.2), but with the first term on the right-hand side replaced by $e^{i\sigma_0 \xi}$, where $\sigma_0 = k_x s$. An equation of this type has been solved by Howe (1995) in a discussion of the production of sound by turbulent flow over infinitely long slots. The corresponding formula for $K_R(k_x, 0, \omega)$ can easily be derived from results given in Howe’s paper, and it is sufficient to quote the final result of this straightforward calculation:

$$K_R(k_x, 0, \omega) = b \mathcal{I}(\sigma_j, b/s), \quad b \gg s, \quad (\text{A } 2)$$

where, in the notation of §3,

$$\begin{aligned} & \mathcal{I}(\sigma_j, b/s) \\ &= \frac{\pi \sigma_+^2 \left(\sum_j \sigma_j [J_0(\sigma_j) - 2W(\sigma_j)] [J_0(\sigma_{j+1})W(\sigma_{j+2}) - J_0(\sigma_{j+2})W(\sigma_{j+1})] \right)}{[(\sigma_+ - \sigma_0)^2 + \mu^2(\sigma_- - \sigma_0)^2] \left[\begin{array}{l} \sigma_1 [J_0(\sigma_1) - 2W(\sigma_1)] [J_0(\sigma_2) - \Psi W(\sigma_2)] \\ - \sigma_2 [J_0(\sigma_2) - 2W(\sigma_2)] [J_0(\sigma_1) - \Psi W(\sigma_1)] \end{array} \right]}, \end{aligned} \quad (\text{A } 3)$$

$\sigma_{\pm} = \omega s / U_{\pm}$, and $\mu = U_- / U_+$. The summation in the numerator is over $j = 0, 1, 2$, with suffixes evaluated modulo 3.

In more general cases, involving apertures of arbitrary shape in grazing flow, a simple extension of the reverse flow reciprocal theorem (Howe 1975; Möhring 1979) can be used to calculate $K_R(\mathbf{k}, \omega)$ from a knowledge of the displacement ζ of the vortex sheet produced by a spatially uniform pressure load $[p]$ when the mean flow direction is reversed. In an ideal fluid of uniform mean density, reverse flow reciprocity implies $\varphi(\mathbf{x}_B, \mathbf{x}_A, \omega) \equiv \varphi^R(\mathbf{x}_A, \mathbf{x}_B, \omega)$ where $\varphi(\mathbf{x}, \mathbf{x}_A, \omega)$, $\varphi^R(\mathbf{x}, \mathbf{x}_B, \omega)$ are the perturbation

velocity potentials produced by equal point sources located respectively at $\mathbf{x}_A$ in the direct problem and at $\mathbf{x}_B$ in the reciprocal problem, provided the mean flow is reversed in the reciprocal problem. The theorem remains valid in the presence of a vortex sheet provided that, in both the direct and reciprocal problems, the sheet is linearly disturbed from its equilibrium position, and the Kutta condition is satisfied. This means that, if γ denotes the edge of the aperture, and γ_S is that portion of γ on which the Kutta condition is satisfied in the direct problem (i.e. where the direction of the mean velocity is directed into the region S occupied by the aperture), then in the reverse flow problem the Kutta condition must be imposed on the conjugate section $\gamma - \gamma_S$ of γ .

The theorem can be applied to show that

$$K_R(\mathbf{k}, \omega) = (2\pi)^2 \rho_0 \omega^2 \zeta^R(-\mathbf{k}) \equiv \rho_0 \omega^2 \int_S \zeta^R(x, y) e^{i\mathbf{k} \cdot \mathbf{x}} dx dy', \quad (\text{A } 4)$$

where $\zeta^R(x, y)$ is the displacement of the vortex sheet when a *spatially uniform* time harmonic pressure load of *unit amplitude* ($[p] \equiv 1$) is applied and the directions of the mean flows above and below the plane are reversed. In this reciprocal problem ζ^R satisfies the Kutta condition over the section $\gamma - \gamma_S$ of γ . Equation (A 4) is obtained by equating Fourier integral representations of the perturbation potentials at $\mathbf{x}_B$ and $\mathbf{x}_A$ generated respectively by point sources at $\mathbf{x}_A$ and $\mathbf{x}_B$ in the direct and reciprocal problems and letting $|\mathbf{x}_B| \rightarrow \infty$, so that the motion of the vortex sheet in the reciprocal problem is ultimately driven by a uniform pressure.

References

- Bechert, D. W. 1979 Sound absorption caused by vorticity shedding, demonstrated with a jet flow. AIAA Paper 79-0575.
- Carrier, G. F., Krook, M. & Pearson, C. E. 1966 *Functions of a complex variable*. New York: McGraw-Hill.
- Crighton, D. G. 1981 Acoustics as a branch of fluid mechanics. *J. Fluid Mech.* **106**, 261–298.
- Crighton, D. G. 1985 The Kutta condition in unsteady flow. *Ann. Rev. Fluid Mech.* **17**, 411–445.
- DeMetz, F. C. & Farabee, T. M. 1977 Laminar and turbulent shear flow induced cavity resonances. AIAA paper 77-1293.
- Howe, M. S. 1975 The generation of sound by aerodynamic sources in an inhomogeneous steady flow. *J. Fluid Mech.* **67**, 579–610.
- Howe, M. S. 1979 On the theory of unsteady high Reynolds number flow through a circular aperture. *Proc. R. Soc. Lond. A* **366**, 205–233.
- Howe, M. S. 1980 On the diffraction of sound by a screen with circular apertures in the presence of a low Mach number grazing flow. *Proc. R. Soc. Lond. A* **370**, 523–544.
- Howe, M. S. 1981a On the theory of unsteady shearing flow over a slot. *Phil. Trans. R. Soc. Lond. A* **303**, 151–180.
- Howe, M. S. 1981b The influence of mean shear on unsteady aperture flow, with application to acoustical diffraction and self-sustained cavity oscillations. *J. Fluid Mech.* **109**, 125–146.
- Howe, M. S. 1995 Influence of mean shear on sound produced by turbulent flow over surface slots. *J. Acoust. Soc. Am.* **97**, 1522–1533.
- Hughes, I. J. & Dowling, A. P. 1990 The absorption of sound by perforated linings. *J. Fluid Mech.* **218**, 299–336.
- Kellogg, O. D. 1954 *Foundations of potential theory*. New York: Dover.
- Lamb, H. 1932 *Hydrodynamics* (6th edn; reprinted 1993). Cambridge University Press.
- Möhring, W. 1979 Modelling low Mach number noise. In *Proc. Symp. Mechanics of Sound Generation in Flows* (Göttingen, August 28–31 1979) (ed. E.-A. Müller), pp. 85–96. Berlin: Springer.

- Rayleigh, Lord 1870 On the theory of resonance. *Phil. Trans. R. Soc. Lond.* **161**, 77–118.
- Rayleigh, Lord 1945 *Theory of sound*, vol. 2. New York: Dover.
- Rockwell, D. 1983 Oscillations of impinging shear layers. *AIAA JI* **21**, 645–664.
- Scott, M. I. 1995 The Rayleigh conductivity of a circular aperture in the presence of a grazing flow. M.S. thesis, Boston University, College of Engineering.
- Sneddon, I. N. 1966 *Mixed boundary value problems in potential theory*. Amsterdam: North Holland.
- Vér, I. L. 1990 Practical examples of noise and vibration control: case history of consulting projects. *Noise Control Engng J.* **35** 115–125.

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